

SolarGrid Demo Utility — Renewable Forecast and Forecast Error Policy

SG-FCST-004

SOLARGRID AI — Synthetic Engineering Demonstration Corpus

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Document Title	SolarGrid Demo Utility — Renewable Forecast and Forecast Error Policy
Version	1.0
Revision Date	2026-01-15
Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY
Purpose	Define synthetic solar forecasting, uncertainty handling, reserve adjustment, and forecast-driven replanning.

Demo Note: This document contains synthetic engineering requirements created for the SOLARGRID AI demonstration. All numerical values, thresholds, equipment ratings, and procedures are illustrative and must be replaced or validated by the deploying organization before operational use.

Use restriction: This document is intended only for RAG testing, semantic retrieval testing, SafetyAgent demonstration, engineering evidence demonstration, PlannerAgent integration testing, and Tool 16 testing. It is not intended for real-world grid operation.

Document Control

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Revision History

Revision	Date	Description	Author
1.0	2026-01-15	Initial synthetic demonstration release	SolarGrid Demo Utility — Engineering Department

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1. Purpose and Scope

This policy defines the synthetic forecasting workflow for the SolarGrid Demo Utility solar PV fleet. It describes day-ahead and intra-day forecasts, uncertainty, forecast error, confidence intervals, reserve adjustment, update frequency, abnormal conditions, and replanning triggers.

Forecast thresholds are configurable demonstration parameters, not weather-service requirements or regulatory rules.

2. Solar Forecast Definitions

FCST-001 — The solar forecast shall represent expected aggregate PV active power in MW for the relevant planning interval.

FCST-002 — Forecast error shall be represented as the difference between realized or latest estimated PV output and the forecast value for the same interval.

Forecast Product	Update Frequency	Primary Use
Day-ahead	Once per planning cycle	Unit commitment and reserve planning
Intra-day	Every 30 minutes	Plan refinement and reserve adjustment
Near-term	Every 5 minutes when enabled	Operational monitoring and short-horizon response

3. Forecast Uncertainty

Forecast uncertainty is represented by an expected value plus an uncertainty band. The demonstration system may use 50%, 80%, and 95% confidence intervals for analysis.

FCST-003 — Reserve planning shall consider forecast uncertainty when the expected solar forecast error could materially affect the operating reserve margin.

FCST-004 — A wider forecast uncertainty band shall not be interpreted as a guaranteed physical output range; it is an input to planning risk analysis.

4. Forecast Error and Reserve Adjustment

FCST-005 — A solar forecast change greater than 15% from the currently approved planning value shall trigger a plan-impact assessment.

FCST-006 — When a material forecast deviation reduces expected generation, the planner shall evaluate whether operating reserve remains at or above 25 MW.

Example: a 120 MW solar forecast is revised to 95 MW. The change is approximately 20.8%, exceeding the synthetic 15% material-change threshold. The system should re-evaluate reserve, generator commitment, BESS dispatch, and network flows.

5. Abnormal Forecast Conditions

FCST-007 — Rapid cloud-cover transitions, unexpected curtailment, sensor quality alarms, or communication loss shall be represented as abnormal forecast conditions.

FCST-008 — When forecast quality is degraded, the planning process shall use the configured conservative uncertainty treatment and may require additional reserve.

6. Forecast Update Workflow

1. Ingest latest PV telemetry and forecast data. 2. Compare the new forecast with the approved baseline. 3. Calculate forecast error and uncertainty indicators. 4. Check the material-change threshold. 5. Recalculate reserve margin if

required. 6. Trigger impact assessment or replanning when criteria are met.

7. Synthetic Examples

Scenario	Old Forecast	New Forecast	Change	Planning Action
Stable solar	100 MW	104 MW	+4%	Monitor; no mandatory replan
Material decrease	120 MW	95 MW	−20.8%	Impact assessment and reserve check
Large increase	70 MW	90 MW	+28.6%	Impact assessment; check curtailment and minimum generation
Sensor uncertainty	110 MW	110 MW	0%	Use uncertainty treatment if quality alarm active

8. RAG Evidence Statements

FCST-009 — A forecast update is evidence about expected renewable output, not direct evidence that the network can safely accommodate the resulting dispatch.

FCST-010 — A material renewable forecast change shall be evaluated together with generator, BESS, reserve, and network evidence before a revised plan is approved.